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GKConv-Guided CP Decomposition for Hyperspectral and Multispectral Image Fusion

Chuangjie Fang, Yan Li, Honghui Xu, *Member, IEEE,* and Jianwei Zheng, *Member, IEEE*

***Abstract*—Hyperspectral and multispectral image fusion (HSI-MSI fusion) aims to reconstruct a high-resolution hyperspectral (HRHS) image from a low-resolution hyperspectral (LRHS) image and a co-registered high-resolution multispectral (HRMS) image. Low-rank priors have demonstrated their effectiveness in this task and have been incorporated into model-guided deep learning frameworks. However, in many existing approaches, such priors are imposed primarily through regularization or optimization-inspired procedures, whereas their direct integration as internal tensor reconstruction mechanisms has received comparatively less attention. Motivated by this observation, we propose the canonical polyadic tensor reconstruction network (CPTRNet), which combines explicit CP-structured low-rank reconstruction with nonlinear factor learning in a differentiable deep framework. Specifically, the proposed rank-1 tensor generation (RTG) module employs a Gram-polynomial-based KAN convolution (GKConv) to extract expressive nonlinear features for factor generation. The Kronecker basis vector generation (KVG) module then aggregates these features into three mode-wise factor vectors, whose outer product produces a rank-1 tensor. By cascading multiple RTG units, the resulting CP tensor reconstruction (CPTR) block aggregates multiple rank-1 components to form a structured low-rank representation. Extensive experiments demonstrate that CPTRNet achieves superior HSI-MSI fusion performance with fewer parameters and lower computational complexity than state-of-the-art methods. The source code is available at https://github.com/chuangjiefang/CPTRNet.**

***Index Terms*—Hyperspectral and multispectral image fusion, CP decomposition, Kolmogorov-Arnold network, low-rank.**

I. Introduction

Hyperspectral images (HSIs) comprise numerous, contiguous, and narrow spectral bands, each capturing the reflectance or radiance properties of the observed surface. Unlike traditional color or multispectral images, HSIs encompass a large number of continuous spectral channels across a broad electromagnetic spectrum. This characteristic enables superior scene representation, thereby benefiting a range of computer vision tasks, such as classification [1], [2], segmentation [3], [4], detection [5]–[7], and object tracking [8], [9]. However, due to the hardware limitations of imaging sensors, traditional optical satellites typically produce either high-resolution multispectral (HRMS) images with limited spectral bands or low-resolution hyperspectral (LRHS) images with lower spatial resolution. Consequently, fusing HRMS and LRHS images to generate high-resolution hyperspectral (HRHS) images has become a widely adopted paradigm to overcome these trade-offs.

In the past decade, HSI-MSI fusion has witnessed significant advancements, with model-based and deep learning (DL)-based methods emerging as the predominant approaches. Within unsupervised HSI-MSI fusion, model-based approaches [10], [11] effectively leverage image priors and estimate the latent HRHS image by modeling the degradation process from the latent HRHS image to the observed LRHS and HRMS images. Among these approaches, tensor-decomposition-based methods [12], [13] provide an important framework for multisource remote sensing data fusion. Most of them decouple spectral features and spatial structures by factorizing high-dimensional spectral data into low-rank tensor components and mode-wise factor matrices. However, conventional tensor decomposition methods are often based on linear factorization, which may limit their ability to characterize complex nonlinear spatial-spectral relationships. Moreover, they commonly rely on iterative optimization, which can incur considerable computational overhead during inference. These limitations, together with the availability of paired training data, have contributed to the increasing adoption of supervised DL-based methods in HSI-MSI fusion.

Owing to their superior feature representation capabilities, DL-based methods [14]–[17] have achieved remarkable progress across diverse domains. In HSI-MSI fusion, convolutional neural networks (CNNs) [18], [19] extract local features through weight-sharing mechanisms, offering notable computational efficiency. However, traditional CNNs typically model spatial dependencies via sequences of linear operations with fixed activation functions, limiting their capacity to represent complex, high-dimensional, and nonlinear relationships. In contrast, the recently proposed Kolmogorov-Arnold network (KAN) convolution [20], [21] replaces the linear weights with B-spline functions, demonstrating promising potential for modeling high-dimensional nonlinear relationships. However, the incorporation of B-spline activations introduces significant computational overhead, prompting the exploration of alternative basis functions. Moreover, similar to traditional CNNs, KAN convolutions remain limited in their capacity to capture long-range dependencies. These limitations collectively hinder their effective deployment in HSI-MSI fusion.

Beyond these model-specific limitations, the inherent "black-box" nature of DL methods limits interpretability and obscures how structural priors contribute to feature reconstruction. Low-rank priors have been shown to effectively characterize the intrinsic structure of high-dimensional remote

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Chuangjie Fang, Yan Li, and Honghui Xu are with the College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou 310023, China (e-mail: fangcj@zjut.edu.cn; lirui@zjut.edu.cn; xhh9609@gmail.com).

Jianwei Zheng is with the College of Computer Science and Technology, Zhejiang University of Technology, Hangzhou 310023, China, and also with the Zhejiang Key Laboratory of Visual Information Intelligent Processing, Hangzhou 310023, China (e-mail: zjw@zjut.edu.cn).

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sensing data and have also been incorporated into model-guided deep learning frameworks [22]–[24]. However, within these frameworks, they are often introduced at the objective or algorithmic level, for example through regularization terms or unfolded optimization steps. Directly embedding tensor factorization into the feature-reconstruction pathway therefore offers a complementary architectural direction. This direction is well aligned with the requirements of HSI reconstruction, which must preserve fine spatial details while exploiting the substantial spectral redundancy of HSIs. Accordingly, a key design consideration is to realize this low-rank structure through an explicit tensor reconstruction mechanism while retaining sufficient nonlinear capacity to represent complex local variations.

Motivated by these observations, we propose the canonical polyadic tensor reconstruction network (CPTRNet), which instantiates canonical polyadic (CP) decomposition as an internal differentiable reconstruction mechanism rather than solely as an external constraint. A key challenge is to preserve the compact spatial-spectral structure induced by CP factorization while providing the factor vectors with sufficient capacity to model complex nonlinear distributions. CPTRNet addresses this challenge by coupling CP-structured low-rank reconstruction with nonlinear factor learning in a single forward pass, thereby avoiding the repeated per-image iterative updates typical of conventional tensor decomposition algorithms.

To implement this design, we develop a rank-1 tensor generation (RTG) module comprising a Gram-polynomial-based KAN convolution (GKConv) module and a Kronecker basis vector generation (KVG) module. By embedding learnable Gram-polynomial basis functions into the KAN convolution framework, GKConv extracts expressive nonlinear features, which KVG aggregates into compact mode-wise factor vectors. The outer product of these vectors produces a rank-1 tensor. A CP tensor reconstruction (CPTR) block then aggregates the outputs of multiple RTG modules into a structured low-rank representation. Notably, this outer-product construction provides global support across tensor modes, enabling factorized long-range coupling among spatial-spectral elements. This factorized representation is well aligned with the spatial-spectral statistics of HSIs, including spectral collinearity and spatial redundancy. Consequently, this design effectively alleviates the limited receptive field of the GKConv operator, allowing the model to capture fine-grained local details while simultaneously modeling global contextual information. The primary contributions of this paper are summarized as follows:

1) Building on CP theory, we develop CPTRNet, where RTG modules generate rank-1 tensors from learned mode-wise factors and CPTR blocks aggregate them into structured low-rank representations. By introducing a CP-structured low-rank inductive bias for efficient redundancy suppression, CPTRNet achieves high-fidelity HSI-MSI fusion while avoiding the computational overhead associated with iterative tensor decomposition.
2) To provide the CP-structured reconstruction with sufficient nonlinear representation capacity, we embed learnable Gram-polynomial basis functions into the KAN convolution framework, resulting in the proposed GKConv. Compared with traditional convolutions, GKConv substantially enhances nonlinear factor learning while preserving translation equivariance and weight sharing.
3) Extensive experiments conducted on five benchmark datasets demonstrate that CPTRNet outperforms state-of-the-art HSI-MSI fusion methods while requiring fewer parameters and incurring lower computational cost.

II. Related Work

A. DL-Based HSI-MSI Fusion

In HSI-MSI fusion, Transformers have garnered significant attention due to their proficiency in extracting global features. The core advantage lies in their ability to dynamically allocate feature weights via self-attention mechanisms, effectively enhancing cross-scale feature interactions. Li et al. [25] proposed a cross-scale non-local attention network tailored for HSI-MSI fusion, which effectively captures non-local spatial-spectral features. Sun et al. [26] introduced a multiscale cross-fusion Transformer network for reconstructing high-resolution HSIs. This network employs multi-head cross-attention modules and progressive reconstruction blocks to extract spatial-spectral features across different dimensions, significantly improving fusion performance. Meanwhile, CNN-based methods [27], [28] extract local features through weight-sharing mechanisms, demonstrating significant advantages in computational efficiency. Ran et al. [29] proposed GuidedNet, a dual-branch CNN framework that leverages multiscale HRMS image guidance to reconstruct HRHS images with enhanced spatial fidelity and preserved spectral consistency. Wang et al. [18] introduced ARConv, an adaptive convolutional module that learns kernel shapes and sampling densities conditioned on object scale, effectively enhancing fusion quality when integrated into the ARNet architecture.

Alongside these predominantly supervised networks, recent studies have explored unsupervised architectures that reduce reliance on paired training data. Su et al. [30] proposed DTUML, which employs a dual-stream encoder with dilated transformations to preserve spectral information from the LRHS image and capture multiscale spatial details from the HRMS image. Xu et al. [31] introduced FusGAT, which combines multiscale convolutions with graph attention to model non-local relationships in the spectral stream and transfer complementary information to the spatial stream. Together, these methods highlight the potential of multiscale and non-local cross-modal representation learning for unsupervised HSI-MSI fusion and offer complementary strategies to jointly model spatial and spectral information. Despite these advances, conventional convolution-based designs may remain limited in modeling complex nonlinear relationships, whereas attention-based mechanisms can introduce considerable computational cost. These considerations motivate more expressive and efficient nonlinear operators for HSI-MSI fusion.

B. KAN Model

Inspired by the Kolmogorov-Arnold theorem, KANs [32] have recently emerged as a new computational architecture.

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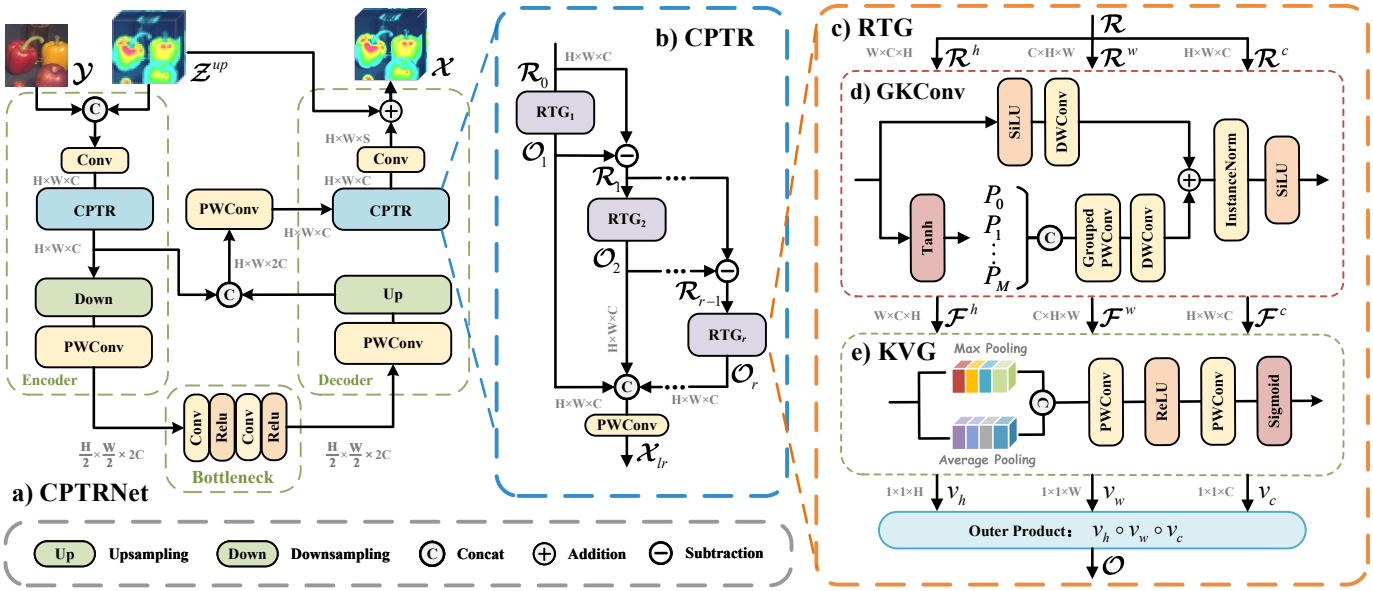


Fig. 1. Overview of CPTRNet. (a) Overall HSI-MSI fusion architecture. (b) CPTR block for progressive residual decomposition and rank-1 component aggregation. (c) Three-branch RTG module for rank-1 tensor generation. (d) GKConv for nonlinear feature extraction. (e) KVG for factor generation.

These networks replace traditional linear weights with learnable univariate functions parameterized by splines, enhancing both interpretability and the ability to capture complex and high-dimensional relationships. Li et al. [33] integrated KAN modules into the U-Net architecture and demonstrated strong performance in medical image segmentation. However, the use of fully connected structures in KAN models leads to a large number of parameters. To address this issue, Bodner et al. [20] proposed a convolutional variant of KAN that replaces the fully connected layers with convolutional operations featuring parameter sharing, thereby effectively reducing the model's parameter count. Zheng et al. [34] further extended this approach to hyperspectral image classification by fully harnessing the complementary strengths of KAN convolutions and GNN architectures, effectively addressing the limitations of MLPs in scalability and interpretability. Nevertheless, the aforementioned KAN convolutions still employ complex B-spline functions, leading to significant computational overhead. In addition, similar to traditional CNNs, they remain limited in their ability to capture long-range dependencies.

C. CP Decomposition

Given that both HSIs and MSIs exhibit 3D structures over spatial and spectral dimensions, tensor decomposition methods [35]–[38] have emerged as natural and efficient approaches for processing such data. Low-rank tensor recovery techniques aim to reconstruct the original data structure from degraded observations by imposing prior constraints. Among these, CP decomposition has gained widespread application in computer vision and image reconstruction due to its effectiveness in capturing the intrinsic low-rank properties of data. This approach decomposes the target tensor into a weighted sum of rank-1 tensors, with each rank-1 component capturing a separable structural pattern of varying importance across modes. Inspired by the CP framework, Xie et al. [39] proposed a Kronecker-based measurement scheme that offers a theoretical interpretation of sparse representations under CP-induced low-rank constraints. Chen et al. [40] further advanced this field by introducing a 3D contextual representation model that improves the practicality and efficiency of tensor decomposition, eliminating the need for traditional spatial compression and simplifying high-order modeling. Subsequently, Zhang et al. [24] extended CP decomposition to snapshot hyperspectral reconstruction by learning a tensor low-rank prior within an iterative deep unfolding framework. However, such deep tensor-prior paradigms are inherently tailored to single-sensor inverse reconstruction. They rely on repeatedly updating rank-1 components under a fixed observation model, resulting in substantial computational overhead. More importantly, their basis generation relies solely on globally pooled features, a design choice that tends to weaken high-frequency spatial cues. While this design is effective for recovering HSIs from compressed measurements, it becomes a major limitation for HSI-MSI fusion, where high-resolution spatial details must be explicitly injected from HRMS observations. These gaps motivate us to develop an end-to-end CP network for HSI-MSI fusion that efficiently couples cross-modal information while improving the expressiveness of the low-rank factors.

III. Methodology

A. Overall Architecture

Given an LRHS image $\boldsymbol{\mathcal{Z}} \in \mathbb{R}^{h\times w\times S}$ with rich spectral information and a co-registered HRMS image $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{H\times W\times s}$ with high spatial resolution captured from the same scene, HSI-MSI fusion aims to reconstruct the target HRHS image $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{H\times W\times S}$. The reconstructed image $\boldsymbol{\mathcal{X}}$ aims to preserve both fine texture details and spectral consistency. Here, $\{H, W, S\}$ represent the spatial dimensions and spectral bands of the target image, where $H > h, W > w$, and $S > s$.

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The overall structure of the CPTRNet architecture is illustrated in Fig. 1. The network takes the LRHS image $\boldsymbol{\mathcal{Z}}$ and the HRMS image $\boldsymbol{\mathcal{Y}}$ as inputs and reconstructs the HRHS image $\boldsymbol{\mathcal{X}}$ as output. Initially, we apply bicubic interpolation to upsample $\boldsymbol{\mathcal{Z}}$ to $\boldsymbol{\mathcal{Z}}^{up} \in \mathbb{R}^{H\times W\times S}$, aligning its spatial dimensions with those of the HRMS image. Subsequently, $\boldsymbol{\mathcal{Z}}^{up}$ and $\boldsymbol{\mathcal{Y}}$ are concatenated, followed by the application of a 3×3 convolutional layer to extract shallow features $\boldsymbol{\mathcal{X}}_0 \in \mathbb{R}^{H\times W\times C}$, where C denotes the number of channels. Afterward, the low-rank feature representation is extracted through an encoder-bottleneck-decoder structure. Both the encoder and decoder consist of a CPTR block and a resizing unit, while the bottleneck is formed by a simple convolutional block designed to enhance the modeling of spectral features, as shown in Fig. 1(a). In each resizing unit, strided or transposed convolutions adjust the spatial resolution, while subsequent PWConv layers facilitate channel-wise interaction and project the features to the channel dimension required by the next stage. Finally, after passing through a 3×3 convolutional layer, the features are added to the global residual to compensate for the loss of detailed features during the encoding-decoding process, resulting in the final output $\boldsymbol{\mathcal{X}}$.

B. CP Decomposition Theory

Given an arbitrary order-N tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$, it can be represented as a linear combination of rank-1 tensors, expressed as:

$$\boldsymbol{\mathcal{T}} = \sum_{i=1}^{r} \omega_i (\boldsymbol{v}_i^{(1)} \circ \boldsymbol{v}_i^{(2)} \circ \cdots \circ \boldsymbol{v}_i^{(N)}), \tag{1}$$

where $\boldsymbol{v}_i^{(n)} \in \mathbb{R}^{I_n}$ denotes the factor vector corresponding to the n-th mode of the i-th rank-1 component (for $n = 1, 2, \ldots, N$), and the term $\boldsymbol{v}_i^{(1)} \circ \boldsymbol{v}_i^{(2)} \circ \cdots \circ \boldsymbol{v}_i^{(N)} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ represents the rank-1 tensor formed by the vector outer product ($\circ$) of these vectors. The weight ω_i corresponds to the coefficient for each rank-1 tensor. The value r denotes the CP rank of the tensor and determines the total number of rank-1 tensors in the decomposition. In the order-3 setting considered in this work, the three modes correspond to height, width, and channel, respectively.

According to Eq. (1), CP decomposition expresses a higher-order tensor as a weighted sum of r rank-1 tensors. In this sense, it can be viewed as a higher-order analogue of matrix low-rank decomposition, where each rank-1 tensor captures a separable structural pattern across different modes. When applied to learned image features, CP decomposition can therefore compactly model structured variations across spatial and spectral modes while naturally imposing a low-rank prior.

C. Rank-1 Tensor Generation (RTG) Module

In CP decomposition, accurate estimation of rank-1 components is critical for tensor reconstruction. However, classical CP decomposition involves a highly nonconvex optimization problem and often suffers from slow convergence, particularly for large or noisy tensors. To address these limitations, we propose the RTG module, as illustrated in Fig. 1(c). The module consists of three parallel branches corresponding to the height, width, and channel dimensions. Given an input residual feature tensor $\boldsymbol{\mathcal{R}} \in \mathbb{R}^{H\times W\times C}$, the three branch inputs, denoted by $\boldsymbol{\mathcal{R}}^h \in \mathbb{R}^{W\times C\times H}$, $\boldsymbol{\mathcal{R}}^w \in \mathbb{R}^{C\times H\times W}$, and $\boldsymbol{\mathcal{R}}^c \in \mathbb{R}^{H\times W\times C}$, are obtained by permuting $\boldsymbol{\mathcal{R}}$ along the height, width, and channel modes, respectively. To precisely capture nonlinear features across these dimensions, each branch incorporates a GKConv submodule to model complex high-dimensional relationships. Subsequently, a KVG submodule aggregates the extracted features into global contextual representations through pooling operations, yielding three mode-wise factor vectors $\boldsymbol{v}^h \in \mathbb{R}^H$, $\boldsymbol{v}^w \in \mathbb{R}^W$, and $\boldsymbol{v}^c \in \mathbb{R}^C$ for the height, width, and channel modes, respectively. These vectors are then combined through the outer product to generate the rank-1 tensor $\boldsymbol{\mathcal{O}} \in \mathbb{R}^{H\times W\times C}$. The generated rank-1 tensor can be written as

$$\boldsymbol{\mathcal{O}} = \boldsymbol{v}^h \circ \boldsymbol{v}^w \circ \boldsymbol{v}^c. \tag{2}$$

By coupling spatial and spectral elements through shared latent factors, the RTG module captures global long-range dependencies, thereby alleviating the limited receptive field of local operators. Compared with self-attention or non-local filtering, this factorized mechanism models global correlations more efficiently, enabling effective feature aggregation and robust low-rank approximation in high-dimensional spaces. Through this factorized coupling, the RTG module provides global contextual support while imposing a separable low-rank inductive bias on the learned representation. To make this algebraic property explicit, Lemma 1 states that the tensor generated by the RTG module satisfies the definition of a rank-1 CP component.

Lemma 1 (Rank-1 Property of the Generated Tensor)**.** *Let $\boldsymbol{v}^h \in \mathbb{R}^H$, $\boldsymbol{v}^w \in \mathbb{R}^W$, and $\boldsymbol{v}^c \in \mathbb{R}^C$ denote the factor vectors generated by the three branches of the RTG module, and let $\boldsymbol{\mathcal{O}} \in \mathbb{R}^{H\times W\times C}$ be the order-3 tensor defined in Eq.* (2)*. Under the unfolding convention in [41], each mode-n unfolding of $\boldsymbol{\mathcal{O}}$ has rank at most one for $n \in \{1, 2, 3\}$. If all three factor vectors are nonzero, then*

$$\operatorname{rank}\big(\boldsymbol{\mathcal{O}}_{(n)}\big) = 1, \qquad n = 1, 2, 3.$$

Moreover,

$$\begin{aligned}
\boldsymbol{\mathcal{O}}_{(1)} &= \boldsymbol{v}^h \big(\boldsymbol{v}^c \otimes \boldsymbol{v}^w\big)^\top, \\
\boldsymbol{\mathcal{O}}_{(2)} &= \boldsymbol{v}^w \big(\boldsymbol{v}^c \otimes \boldsymbol{v}^h\big)^\top, \\
\boldsymbol{\mathcal{O}}_{(3)} &= \boldsymbol{v}^c \big(\boldsymbol{v}^w \otimes \boldsymbol{v}^h\big)^\top,
\end{aligned}$$

where $\otimes$ denotes the Kronecker product.

Proof. By construction, $\boldsymbol{\mathcal{O}}$ is the outer product of three vectors. Under the adopted matricization convention, the mode-n unfoldings take the forms above. Each unfolding is therefore the outer product of a column vector and a row vector, and thus has rank at most one. When $\boldsymbol{v}^h$, $\boldsymbol{v}^w$, and $\boldsymbol{v}^c$ are all nonzero, both factors in each outer product are nonzero, which implies $\operatorname{rank}(\boldsymbol{\mathcal{O}}_{(n)}) = 1$ for $n = 1, 2, 3$. □

Lemma 1 establishes the rank-1 structural property of the tensor generated by a single RTG module. Building on this

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result, aggregating r RTG-generated rank-1 components enlarges the corresponding representational family. The following proposition formalizes this observation by showing that the best achievable approximation error over this family is non-increasing with r.

Proposition 2 (Effect of Using r RTG Modules)**.** *Let $\mathcal{S}_r$ denote the family of order-3 tensors that can be expressed as*

$$\sum_{i=1}^{r} \boldsymbol{v}_i^h \circ \boldsymbol{v}_i^w \circ \boldsymbol{v}_i^c,$$

that is, as the sum of at most r rank-1 CP components, where scalar weights are absorbed into one factor without loss of generality. Since $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$, the best achievable approximation error over these representational families is non-increasing in r. That is, for any target tensor $\boldsymbol{\mathcal{T}}$,

$$\inf_{\widehat{\boldsymbol{\mathcal{T}}} \in \mathcal{S}_{r+1}} \|\boldsymbol{\mathcal{T}} - \widehat{\boldsymbol{\mathcal{T}}}\|_F \leq \inf_{\widehat{\boldsymbol{\mathcal{T}}} \in \mathcal{S}_r} \|\boldsymbol{\mathcal{T}} - \widehat{\boldsymbol{\mathcal{T}}}\|_F.$$

To examine whether this structural inductive bias is reflected in the learned representations, we analyze the singular-value spectra of the feature maps in Fig. 2. As shown in Fig. 2(a), the RTG output exhibits a sharp drop after the first singular value, in contrast to the gradual spectral decay observed for the standard CNN and Transformer blocks. This behavior is further supported by the cumulative explained-energy curves in Fig. 2(b), where the leading singular component of the RTG output captures nearly all the energy. Together, these results confirm that the learned RTG outputs exhibit the rank-1 structure imposed by the outer-product construction.

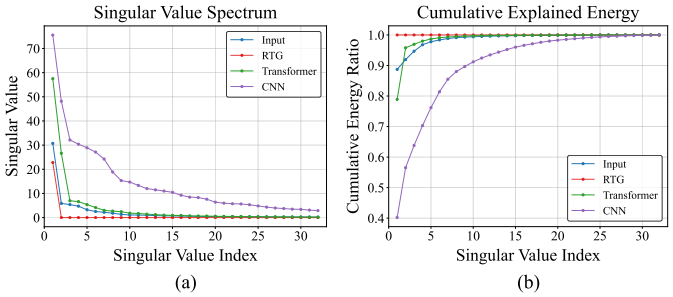


Fig. 2. Singular-value analysis of the input feature and the outputs of different blocks on the CAVE dataset at the $\times 8$ scale. (a) Singular-value spectra, showing the decay of the ordered singular values. (b) Cumulative explained-energy curves, showing the energy captured by the leading singular components. The RTG output exhibits a sharp decay after the first singular value and concentrates nearly all its energy in the leading component, consistent with the rank-1 structure expected from its outer-product construction.

D. CP Tensor Reconstruction (CPTR) Block

Recall that CP decomposition expresses a 3D hyperspectral feature tensor as a sum of r rank-1 components. Based on RTG, the CPTR block extracts these components through progressive residual decomposition and then fuses them to reconstruct the low-rank representation, as illustrated in Fig. 1(b). Given the initial residual feature tensor $\boldsymbol{\mathcal{R}}_0 \in \mathbb{R}^{H\times W\times C}$ as the input to the CPTR block, the first RTG generates the first CP component $\boldsymbol{\mathcal{O}}_1$. Each subsequent RTG is applied to the residual obtained by subtracting the previously estimated components, which encourages later branches to capture information not explained by earlier ones. In practice, this progressive residual decomposition can encourage the separation of complementary spatial-spectral structures, with earlier components tending to capture relatively coarse patterns and later ones refining finer details. Accordingly, the progressive residual decomposition is defined recursively as

$$\begin{aligned} \boldsymbol{\mathcal{O}}_i &= \mathrm{RTG}_i\left(\boldsymbol{\mathcal{R}}_{i-1}\right), \\ \boldsymbol{\mathcal{R}}_i &= \boldsymbol{\mathcal{R}}_{i-1} - \boldsymbol{\mathcal{O}}_i, \qquad i = 1, \ldots, r. \end{aligned} \tag{3}$$

Equivalently, $\boldsymbol{\mathcal{R}}_{i-1} = \boldsymbol{\mathcal{R}}_0 - \sum_{j=1}^{i-1} \boldsymbol{\mathcal{O}}_j$.

Since each RTG outputs a rank-1 tensor as in Eq. (2), the i-th component can be expressed as $\boldsymbol{\mathcal{O}}_i = \boldsymbol{v}_i^h \circ \boldsymbol{v}_i^w \circ \boldsymbol{v}_i^c$. Given the predefined CP rank r, the CPTR block thus generates r distinct rank-1 tensors, each capturing complementary spatial-spectral information. Subsequently, these rank-1 tensors are concatenated and aggregated via a PWConv layer, whose kernel is denoted by K_{agg}. Mathematically, this linear aggregation can be viewed as a generalized weighting operation over the rank-1 tensors, extending the standard scalar weighting in CP decomposition. Consequently, building on the structural guarantee of the RTG module, the recovered representation $\boldsymbol{\mathcal{X}}_{lr}$ is obtained by aggregating r transformed rank-1 components. This preserves a CP-inspired sum-of-components structure with r transformed rank-1 terms. This formulation can be written as

$$\boldsymbol{\mathcal{X}}_{lr} = K_{\mathrm{agg}} * \mathrm{Concat}_{i=1}^{r}[\boldsymbol{\mathcal{O}}_i]_c, \tag{4}$$

where $*$ represents the convolution operation, and $\mathrm{Concat}[\cdot]_c$ indicates channel-wise concatenation.

It is worth noting that traditional CP decomposition aggregates rank-1 tensors through weighted summation. This means that each rank-1 tensor is scaled by a single coefficient, which limits the flexibility of feature modulation across channels. In contrast, we employ a PWConv layer to learn distinct channel-wise mixing weights, leading to a more flexible aggregation. Since the learned tensor encodes rich structural information, it can be viewed as a 3D feature map that models global correlations across different dimensions.

Building upon the canonical CP decomposition in Eq. (1), we substitute the rank-1 form in Eq. (2) into Eq. (4). Since a PWConv layer acting on channel-wise concatenated features can be decomposed into the sum of its sub-kernels applied to the corresponding channel blocks, the CPTR reconstruction can be equivalently written as

$$\begin{aligned} \boldsymbol{\mathcal{X}}_{lr} &= K_{\mathrm{agg}} * \left(\mathrm{Concat}_{i=1}^{r}\left[\boldsymbol{v}_i^h \circ \boldsymbol{v}_i^w \circ \boldsymbol{v}_i^c\right]_c\right) \\ &= \sum_{i=1}^{r} K_{\mathrm{agg},i} * \left(\boldsymbol{v}_i^h \circ \boldsymbol{v}_i^w \circ \boldsymbol{v}_i^c\right), \end{aligned} \tag{5}$$

where $K_{\mathrm{agg},i}$ denotes the sub-kernel of the full PWConv layer corresponding to the i-th concatenated rank-1 component. Thus, compared with the classical CP form in Eq. (1), Eq. (5) retains the sum-of-rank-1-components structure while replacing the scalar coefficients with learnable PWConv transformations. Fig. 3 illustrates this CP-structured representation, where $\boldsymbol{v}_i^h$, $\boldsymbol{v}_i^w$, and $\boldsymbol{v}_i^c$ denote the factor vectors associated with

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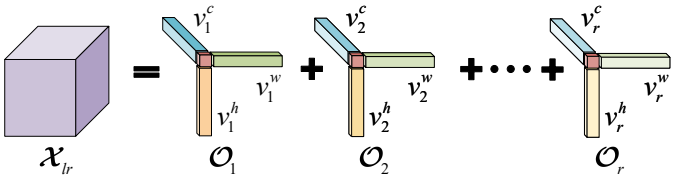


Fig. 3. Illustration of the CP-structured low-rank representation in CPTRNet. Each rank-1 component $\boldsymbol{\mathcal{O}}_i$ is formed from the spatial-height, spatial-width, and feature-channel factors $\boldsymbol{v}_i^h$, $\boldsymbol{v}_i^w$, and $\boldsymbol{v}_i^c$, respectively. The PWConv aggregation K_{agg} is omitted for clarity.

the spatial-height, spatial-width, and feature-channel modes, respectively. Algorithm 1 summarizes the CPTR forward pass, where each RTG uses GKConv and KVG to generate a rank-1 tensor from the current residual.

E. Gram-Polynomial-Based KAN Convolution (GKConv)

To better model nonlinear feature interactions within each RTG branch, we further introduce the Gram-polynomial-based KAN convolution (GKConv), as illustrated in Fig. 1(d). This design is motivated by the Kolmogorov-Arnold theorem, which states that any multivariate continuous function can be expressed as a finite superposition of univariate continuous functions and addition operations. This provides theoretical support for KANs, endowing them with interpretability that is typically absent in general deep networks. In general, a KAN with L nested layers can be written as

$$\mathrm{KAN}(X) = (\Psi_{L-1} \circ \Psi_{L-2} \circ \cdots \circ \Psi_0)(X), \tag{6}$$

where Ψ_ℓ denotes the mapping of the ℓ-th layer and X is the input. Let d_{in} and d_{out} denote the input and output dimensions of a KAN layer, respectively. Then, Ψ_ℓ can be represented as a $d_{\text{out}} \times d_{\text{in}}$ matrix of learnable univariate B-spline functions:

$$\Psi_\ell = \begin{bmatrix} \psi_{\ell,1,1}(\cdot) & \psi_{\ell,1,2}(\cdot) & \cdots & \psi_{\ell,1,d_{\text{in}}}(\cdot) \\ \psi_{\ell,2,1}(\cdot) & \psi_{\ell,2,2}(\cdot) & \cdots & \psi_{\ell,2,d_{\text{in}}}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{\ell,d_{\text{out}},1}(\cdot) & \psi_{\ell,d_{\text{out}},2}(\cdot) & \cdots & \psi_{\ell,d_{\text{out}},d_{\text{in}}}(\cdot) \end{bmatrix}, \tag{7}$$

and the layer output is given by $X_{\ell+1} = \Psi_\ell(X_\ell)$, where each entry $\psi_{\ell,\alpha,\beta}$ acts on the β-th element of X_ℓ, and the resulting responses are summed over β to produce the α-th output element.

By replacing linear weights with learnable univariate nonlinear functions, KAN effectively captures the intrinsic relationships present in visual data. However, the piecewise B-spline formulation and knot interval localization in the original KAN introduce considerable computational overhead, limiting its practical use for reconstructing HRHS images. To overcome this limitation, we employ Gram polynomials as the nonlinear basis in place of splines. This design retains the nonlinear modeling capacity of KAN while markedly reducing computational complexity and enhancing both training and inference efficiency in HSI-MSI fusion.

Specifically, we replace the spline functions in the standard KAN with Gram functions to construct a nonlinear Gram-polynomial basis function ϕ, expressed as follows:

$$\phi(x) = w_b \mathrm{SiLU}(x) + w_s \mathrm{Gram}(x), \tag{8}$$

Algorithm 1: Pseudocode of the CPTR Block

Input: Initial residual feature tensor $\boldsymbol{\mathcal{R}}_0$, CP rank r, Gram-polynomial order M
Output: Low-rank reconstructed feature tensor $\boldsymbol{\mathcal{X}}_{lr}$

```
for i = 1 to r do
    R^h_{i-1}, R^w_{i-1}, and R^c_{i-1} ⇐ Permute R_{i-1} along the height, width, and channel modes
    F^h_i, F^w_i, and F^c_i ⇐ Use the order-M GKConv module in Eq. (17) to extract branch features
    v^h_i, v^w_i, and v^c_i ⇐ Use the KVG module in Eqs. (18) and (19) to obtain factor vectors
    O_i ⇐ Use Eq. (2) to form the RTG rank-1 output
    R_i ⇐ Use Eq. (3) to update the CPTR residual
end
X_lr ⇐ Use Eq. (4) to aggregate all rank-1 tensors
return X_lr
```

where w_b and w_s are weighting coefficients. $\mathrm{SiLU}(x) = x/(1+e^{-x})$, and $\mathrm{Gram}(x) = \sum_{m=0}^{M} c_m P_m(x)$, where c_m are learnable coefficients, and $\{P_m(\cdot)\}_{m=0}^{M}$ are Gram-type polynomials generated by the three-term recurrence:

$$\begin{aligned} P_0(x) &= 1, \\ P_1(x) &= x, \\ P_m(x) &= x\,P_{m-1}(x) - \beta_{m-1,m}\,P_{m-2}(x), \qquad m \geq 2, \end{aligned} \tag{9}$$

where $\beta_{m-1,m}$ denotes learnable coefficients that are shared across all channels within a layer to maintain a unified basis space. In practice, the polynomial branch is evaluated on $\tanh(x)$ to map the input into the interval $[-1, 1]$, which enhances the numerical stability of the Gram-polynomial computations. To balance representational capacity and numerical robustness, the expansion order of the Gram polynomials is set to $M = 3$. This configuration provides a sufficient degree of nonlinearity to capture intricate hyperspectral features while effectively suppressing the numerical oscillations typically associated with higher-order terms. Eq. (8) gives the generic scalar Gram activation used to replace the spline functions in standard KAN. Subsequently, the resulting polynomial bases are concatenated along the channel dimension and reweighted by a channel-wise grouped PWConv with C groups. The overall Gram function can be formulated as a weighted combination of multi-order polynomial bases:

$$\begin{aligned} \mathrm{Gram}(X) &= K_{\text{mix}} * \mathrm{Concat}_{m=0}^{M}[P_m(\tanh(X))]_c \\ &= \sum_{m=0}^{M} K_{\text{mix},m} * P_m(\tanh(X)), \end{aligned} \tag{10}$$

where K_{mix} denotes a grouped 1×1 convolutional kernel applied to the concatenated polynomial features, and $K_{\text{mix},m}$ represents the weights associated with the m-th polynomial basis. Notably, K_{mix} is implemented as a channel-wise grouped 1×1 convolution. This design allows the module to adaptively reweight the expanded basis functions according to the unique spectral characteristics of different bands, thereby enhancing the model's capacity to represent complex, high-dimensional nonlinear relationships in hyperspectral data.

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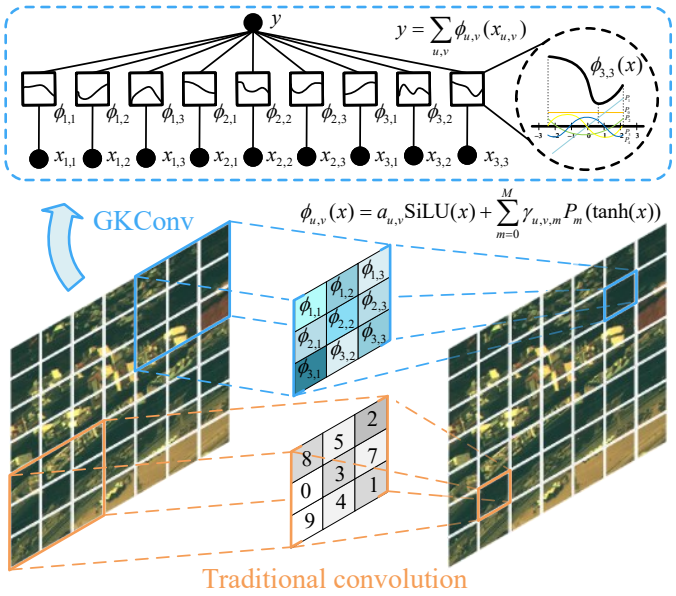


Fig. 4. Comparison of GKConv and traditional convolution. Traditional convolution uses a linear weight at each kernel position, whereas GKConv employs the learnable nonlinear function $\phi_{u,v}(x)$ defined in Eq. (12).

Compared with B-spline functions, Gram polynomials offer enhanced computational and numerical properties. Their three-term recurrence enables a concise implementation with linear computational complexity, reducing computational cost. In addition, the approximately orthogonal basis functions reduce coupling among basis responses, thereby improving numerical stability.

However, the number of activation functions in Gram-polynomial-based KANs still increases with the input and output dimensions, leading to an increased parameter scale and additional computational and memory overhead, which becomes especially significant in high-dimensional hyperspectral tasks. To enable efficient nonlinear spatial modeling, we formulate the Gram-polynomial-based mapping as a channel-wise convolutional operator, referred to as GKConv. Unlike traditional convolution operations in which each kernel element is a linear weight, GKConv assigns a learnable nonlinear activation function to every kernel element, as illustrated in Fig. 4. This design preserves key properties such as weight-sharing and translation equivariance, while substantially improving nonlinear representational capacity. Hence, for each channel, a 3×3 GKConv kernel can be interpreted as a KAN-like layer with 9 input neurons and 1 output neuron. The kernel Φ is expressed as follows:

$$\Phi = \begin{bmatrix} \phi_{11}(\cdot) & \phi_{12}(\cdot) & \phi_{13}(\cdot) \\ \phi_{21}(\cdot) & \phi_{22}(\cdot) & \phi_{23}(\cdot) \\ \phi_{31}(\cdot) & \phi_{32}(\cdot) & \phi_{33}(\cdot) \end{bmatrix}. \tag{11}$$

In GKConv, the number of activation functions grows linearly with the channel dimension. Therefore, for HSI-MSI fusion, GKConv significantly reduces the number of parameters and computational load compared to traditional KANs.

Specifically, for a spatial location $\mathbf{q} \in \{1,\dots,H\} \times \{1,\dots,W\}$, we write $X(\mathbf{q}) \in \mathbb{R}^C$ for the C-dimensional vector at $\mathbf{q}$. For a $k \times k$ kernel, we define the spatial offset set $\Omega_k = \{(u,v) \mid u,v \in \{-\lfloor k/2 \rfloor, \dots, \lfloor k/2 \rfloor\}\}$, which represents the set of relative coordinates within the local receptive field centered at $\mathbf{q}$. Here, $\lfloor\cdot\rfloor$ denotes the floor function, and the basis-enhanced kernel response is defined as:

$$\phi_{u,v}(x) = a_{u,v}\,\mathrm{SiLU}(x) + \sum_{m=0}^{M} \gamma_{u,v,m}\,P_m(\tanh(x)), \tag{12}$$

where x denotes a scalar channel value, and $a_{u,v}$ and $\gamma_{u,v,m}$ are learnable coefficients. The resulting GKConv response at location $\mathbf{q}$, which remains a C-dimensional vector, is then given by

$$\big[\mathrm{GKConv}(X)(\mathbf{q})\big]_c = \sum_{(u,v)\in\Omega_k} \phi_{c,u,v}\big(X_c(\mathbf{q}+(u,v))\big). \tag{13}$$

Substituting Eq. (12) into Eq. (13) and regrouping terms yields:

$$\begin{aligned}\big[\mathrm{GKConv}(X)(\mathbf{q})\big]_c = &\sum_{(u,v)\in\Omega_k} a_{c,u,v}\,\mathrm{SiLU}\big(X_c(\mathbf{q}+(u,v))\big) \\ &+ \sum_{m=0}^{M} \sum_{(u,v)\in\Omega_k} \gamma_{c,u,v,m}\,P_m\big(\tanh(X_c(\mathbf{q}+(u,v)))\big).\end{aligned} \tag{14}$$

Defining the channel-wise convolution kernels $(K_{\mathrm{base}})_c[u,v] = a_{c,u,v}$ and $(K_{\mathrm{poly},m})_c[u,v] = \gamma_{c,u,v,m}$ gives the sum-of-convolutions form:

$$\begin{aligned}\mathrm{GKConv}(X) = &K_{\mathrm{base}} * \mathrm{SiLU}(X) \\ &+ \sum_{m=0}^{M} \big(K_{\mathrm{poly},m} * P_m(\tanh(X))\big),\end{aligned} \tag{15}$$

where K_{base} is the depthwise spatial kernel for the SiLU branch, and $K_{\mathrm{poly},m}$ is the channel-wise spatial kernel for the m-th polynomial basis branch. To reduce the parameter redundancy inherent in modeling each polynomial basis with an independent spatial kernel, the polynomial branch first applies the order-specific grouped 1×1 PWConv kernel $K_{\mathrm{mix},m}$, followed by a shared 3×3 DWConv kernel K_{sp}. Accordingly, under this factorized parameterization, we have:

$$\begin{aligned}&\sum_{m=0}^{M} (K_{\mathrm{poly},m} * P_m(\tanh(X))) \\ =&K_{\mathrm{sp}} * \Big(\sum_{m=0}^{M} K_{\mathrm{mix},m} * P_m(\tanh(X))\Big) \\ =&K_{\mathrm{sp}} * \mathrm{Gram}(X),\end{aligned} \tag{16}$$

which yields the compact formulation of GKConv:

$$\mathrm{GKConv}(X) = K_{\mathrm{base}} * \mathrm{SiLU}(X) + K_{\mathrm{sp}} * \mathrm{Gram}(X). \tag{17}$$

Each kernel element in GKConv employs a dual-path nonlinear activation function. One path uses the SiLU activation to preserve basic nonlinear transformations, while the other constructs an approximately orthogonal basis space using

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Gram polynomials for refined feature fitting. This design maintains the translational equivariance and weight-sharing properties of traditional convolution, while the learnable Gram-polynomial coefficients adaptively combine the basis functions, substantially enhancing the representational capacity for high-dimensional hyperspectral features.

F. Kronecker Basis Vector Generation (KVG)

The key to the RTG module lies in learning informative factor vectors along different modes to faithfully characterize the underlying low-rank structure. In this work, the KVG module is designed to generate three mode-wise factor vectors for subsequent rank-1 tensor construction. Inspired by channel attention mechanisms, we design the KVG module, which incorporates both adaptive average pooling and adaptive max pooling to capture complementary responses along three different modes, as illustrated in Fig. 1(e).

Specifically, for the channel branch, let the input feature tensor be denoted by $\boldsymbol{\mathcal{F}}^c \in \mathbb{R}^{H\times W\times C}$. Spatial average pooling and max pooling are then applied to obtain two 1-D pooled descriptors, $\boldsymbol{p}_{\text{avg}}$ and $\boldsymbol{p}_{\max} \in \mathbb{R}^C$, which summarize complementary global responses:

$$\boldsymbol{p}_{\text{avg}} = \text{AveragePool}(\boldsymbol{\mathcal{F}}^c), \quad \boldsymbol{p}_{\max} = \text{MaxPool}(\boldsymbol{\mathcal{F}}^c). \tag{18}$$

Subsequently, $\boldsymbol{p}_{\text{avg}}$ and $\boldsymbol{p}_{\max}$ are concatenated to form a pooled descriptor $\boldsymbol{p} \in \mathbb{R}^{2C}$. We then employ two PWConv layers with ReLU and Sigmoid activations to perform a nonlinear projection of the pooled descriptor, yielding the channel-mode factor vector $\boldsymbol{v}^c \in \mathbb{R}^C$.

$$\begin{aligned} \boldsymbol{p} &= \text{Concat}(\boldsymbol{p}_{\text{avg}}, \boldsymbol{p}_{\max}), \\ \boldsymbol{v}^c &= \text{Sigmoid}\Big(\text{PWConv}\big(\text{ReLU}\big(\text{PWConv}(\boldsymbol{p})\big)\big)\Big). \end{aligned} \tag{19}$$

The resulting vector $\boldsymbol{v}^c$ serves as the factor vector for the channel mode. For the width and height branches, the same operations are applied to $\boldsymbol{\mathcal{F}}^w \in \mathbb{R}^{C\times H\times W}$ and $\boldsymbol{\mathcal{F}}^h \in \mathbb{R}^{W\times C\times H}$ to obtain $\boldsymbol{v}^w \in \mathbb{R}^W$ and $\boldsymbol{v}^h \in \mathbb{R}^H$, respectively. The KVG module fully exploits the information compression capability of pooling operations, enabling more accurate extraction of factor vectors from high-dimensional feature spaces.

IV. Experiments

A. Experimental Settings

To validate the effectiveness of the proposed method, we conducted experiments on 4 publicly available benchmark datasets: CAVE [53], Houston 2018 [54], Chikusei [55], and Pavia Centre [56]. In addition, we used the real-world WHU-MHF [57] dataset to assess the practical applicability of the proposed method in satellite remote sensing scenarios. Furthermore, to demonstrate the utility of the reconstructed images in downstream analysis, we conducted a hyperspectral image classification experiment on the Pavia Centre dataset.

The CAVE dataset consists of 32 indoor HSIs, each with spatial dimensions of 512×512 pixels and 31 spectral bands ranging from 400 to 700 nm. Of these, 20 images were used for training and the remaining 12 for testing. The Houston 2018 dataset provides an airborne HSI covering the University of Houston campus and the surrounding urban area. The HSI comprises 48 spectral bands covering wavelengths from 380 to 1050 nm and has spatial dimensions of 1202×4172 pixels at a ground sampling distance (GSD) of 1 m. We partitioned the image into 16 non-overlapping 512×512 patches, using 12 for training and 4 for testing. The Chikusei dataset was collected in Ibaraki Prefecture, Japan, and depicts agricultural and urban areas. The HSI has spatial dimensions of 2517×2335 pixels and contains 128 spectral bands spanning wavelengths from 363 to 1018 nm. We extracted 16 non-overlapping 512×512 patches from the dataset, of which 10 were used for training and 6 for testing. The Pavia Centre dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) during a flight over Pavia in northern Italy. The HSI has spatial dimensions of 1096×1096 pixels and comprises 102 spectral bands covering wavelengths from 430 to 860 nm. In our experiments, 8 256×256 patches were cropped from the upper-left region of the HSI for training, and 4 additional non-overlapping 256×256 patches were selected from the remaining area for testing. The WHU-MHF dataset, acquired by the ZY-1-02D satellite, consists of 6 spaceborne LRHS-HRMS image pairs. The LRHS images have a GSD of 30 m and comprise 148 spectral bands, whereas the HRMS images have a GSD of 10 m and comprise 8 spectral bands. In our experiments, 5 image pairs were used for training and 1 for testing.

B. Implementation Details

1) Training Details: To generate LRHS images, an $a \times a$ Gaussian filter with a standard deviation of 2 was applied to blur the HRHS images, followed by spatial downsampling by a factor of a. HRMS images were derived from the HRHS images through spectral degradation. During training, we randomly extracted HRHS patches of size 64×64, from which the corresponding LRHS patches of size $(64/a) \times (64/a)$ and HRMS patches of size 64×64 were generated using the degradation process described above. Data augmentation, including geometric flipping and rotation, was employed to generate additional training samples.

For the WHU-MHF dataset, we adopted a reduced-resolution training strategy due to the absence of ground truth (GT). The network was trained on downsampled image pairs with resolutions of 90 m and 30 m, derived from the original data. We constructed the training set by extracting overlapping patches, with the LRHS and HRMS patch sizes set to 40×40 and 120×120, respectively. This process yielded a total of 378 image pairs for training. Finally, inference was performed directly on the original 30 m LRHS and 10 m HRMS inputs.

2) Parameter Settings: To ensure fairness and reproducibility, we strictly followed the experimental configurations specified by the original authors for all competing methods. All experiments were conducted in a consistent computational environment using the PyTorch framework on a single NVIDIA RTX 3090 GPU. Our model was trained using the Adam optimizer with an initial learning rate of 3×10^{-4}, which was decayed by a factor of 0.95 every 5 epochs during the

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TABLE I
AVERAGE QUANTITATIVE RESULTS OF ALL METHODS ON FOUR DATASETS AT THE $\times 4$ AND $\times 8$ SCALES. THE BEST AND SECOND-BEST VALUES ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY. THE MODEL PARAMS AND FLOPS ARE MEASURED ON THE CAVE DATASET. "M" AND "G" DENOTE MILLIONS AND BILLIONS, RESPECTIVELY.

Scale	Method	CAVE				Houston				Chikusei				Pavia Centre				Params	FLOPs
		PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	(M)	(G)
×4	HySure$_{15}$ [42]	43.12	0.9765	6.780	2.318	45.13	0.9820	2.592	2.198	33.22	0.8864	4.192	5.284	43.95	0.9685	4.855	2.412	—	—
	GTNN$_{25}$ [43]	46.92	0.9898	4.383	1.318	48.01	0.9832	1.907	1.736	37.42	0.9410	3.193	3.822	45.90	0.9775	3.611	1.868	—	—
	GNLTN$_{25}$ [44]	47.08	0.9908	4.025	1.273	48.52	0.9860	1.779	1.511	36.47	0.9355	3.412	4.020	46.12	0.9781	3.578	1.826	—	—
	FusGAT$_{25}$ [31]	46.75	0.9883	4.551	1.382	48.72	0.9876	1.680	1.452	37.27	0.9390	3.107	3.761	46.05	0.9793	3.439	1.861	—	—
	DTUML$_{25}$ [30]	49.85	0.9959	2.181	0.987	50.41	0.9927	1.366	1.074	42.49	0.9673	2.724	3.191	46.69	0.9807	3.317	1.780	—	—
	DTLP$_{21}$ [24]	51.13	0.9966	2.047	0.847	52.64	0.9965	1.047	0.768	42.35	0.9600	2.725	2.315	48.40	0.9840	3.068	1.626	2.82	9.82
	DHIF$_{22}$ [45]	52.28	0.9973	1.840	0.746	53.05	0.9968	0.993	0.721	43.46	0.9660	2.682	2.093	48.09	0.9838	3.124	1.731	22.67	56.50
	MSST$_{23}$ [46]	51.15	0.9968	2.062	0.832	51.59	0.9956	1.146	0.886	43.30	0.9654	2.707	2.196	48.23	0.9835	3.136	1.671	34.39	46.45
	DSPNet$_{23}$ [47]	51.79	0.9970	2.008	0.793	52.43	0.9963	1.025	0.769	41.86	0.9574	2.732	2.396	48.25	0.9836	3.103	1.642	6.06	6.60
	MIMO$_{24}$ [48]	51.49	0.9967	2.083	0.834	52.48	0.9963	1.036	0.763	42.34	0.9585	2.720	2.281	49.06	0.9844	3.020	1.596	4.98	6.53
	KNLNet$_{24}$ [49]	51.82	0.9969	2.014	0.800	52.81	0.9966	0.992	0.738	43.19	0.9637	2.647	2.135	49.02	0.9847	3.007	1.599	1.52	23.60
	DCT$_{24}$ [50]	51.48	0.9968	2.065	0.827	53.72	0.9972	0.945	0.681	43.78	0.9693	2.604	2.079	48.98	0.9846	3.003	1.599	8.14	76.79
	DCINN$_{24}$ [51]	52.68	0.9974	1.831	0.721	53.65	0.9971	0.933	0.678	42.49	0.9612	2.652	2.238	48.53	0.9841	3.072	1.628	5.00	104.15
	CYformer$_{25}$ [52]	52.48	0.9973	1.847	0.742	53.77	0.9971	0.927	0.671	43.65	0.9701	2.621	2.167	49.05	0.9846	3.039	1.610	2.10	5.82
	CPTRNet	**53.03**	**0.9975**	**1.724**	**0.689**	**54.32**	**0.9976**	**0.867**	**0.620**	**44.27**	**0.9714**	**2.539**	**2.002**	**49.46**	**0.9852**	**2.957**	**1.575**	**1.43**	**3.99**
×8	HySure$_{15}$ [42]	40.27	0.9712	8.167	1.537	41.31	0.9598	4.112	1.708	30.61	0.8369	6.742	4.812	41.26	0.9612	5.852	1.552	—	—
	GTNN$_{25}$ [43]	45.06	0.9875	4.621	0.801	43.07	0.9712	3.268	1.502	32.35	0.9095	4.253	3.693	43.73	0.9731	4.055	1.081	—	—
	GNLTN$_{25}$ [44]	45.61	0.9881	4.596	0.699	44.44	0.9751	2.525	1.365	32.52	0.9120	4.247	3.812	44.98	0.9761	3.823	0.967	—	—
	FusGAT$_{25}$ [31]	44.51	0.9856	4.931	0.914	44.42	0.9767	2.502	1.297	33.27	0.9239	3.976	3.384	44.60	0.9763	3.802	1.041	—	—
	DTUML$_{25}$ [30]	46.93	0.9935	2.873	0.689	46.23	0.9896	1.892	0.861	40.17	0.9362	3.029	2.050	45.35	0.9789	3.697	1.017	—	—
	DTLP$_{21}$ [24]	48.31	0.9945	2.585	0.592	48.63	0.9933	1.516	0.603	39.08	0.9315	3.248	1.594	48.02	0.9834	3.145	0.832	2.82	9.82
	DHIF$_{22}$ [45]	50.19	0.9961	2.119	0.484	49.17	0.9936	1.428	0.558	40.11	0.9358	3.092	1.435	47.47	0.9818	3.343	1.088	22.67	56.50
	MSST$_{23}$ [46]	49.08	0.9953	2.302	0.541	48.46	0.9926	1.539	0.618	37.64	0.9243	3.522	1.916	47.96	0.9833	3.152	0.837	34.39	46.45
	DSPNet$_{23}$ [47]	50.28	0.9959	2.194	0.489	48.96	0.9934	1.430	0.567	39.35	0.9318	3.083	1.540	47.97	0.9833	3.151	0.832	6.06	6.60
	MIMO$_{24}$ [48]	49.04	0.9948	2.579	0.559	48.24	0.9929	1.538	0.610	38.77	0.9312	3.129	1.613	48.28	0.9835	3.137	0.837	4.98	6.53
	KNLNet$_{24}$ [49]	48.97	0.9950	2.470	0.564	48.87	0.9935	1.436	0.574	40.29	0.9372	3.043	1.416	48.37	0.9840	3.095	0.830	1.52	23.60
	DCT$_{24}$ [50]	50.50	0.9962	2.118	0.472	49.61	0.9943	1.366	0.534	40.42	0.9418	3.035	1.409	48.50	0.9840	3.090	0.822	8.14	76.79
	DCINN$_{24}$ [51]	50.54	0.9961	2.159	0.477	49.58	0.9943	1.362	0.530	40.16	0.9388	2.997	1.410	48.48	0.9839	3.095	0.840	5.00	104.15
	CYformer$_{25}$ [52]	50.45	0.9960	2.172	0.481	49.85	0.9945	1.341	0.520	40.49	0.9423	3.012	1.398	48.40	0.9839	3.092	0.835	2.10	5.82
	CPTRNet	**50.80**	**0.9963**	**2.034**	**0.454**	**50.30**	**0.9950**	**1.257**	**0.490**	**40.99**	**0.9483**	**2.894**	**1.341**	**48.92**	**0.9844**	**3.048**	**0.821**	**1.43**	**3.99**

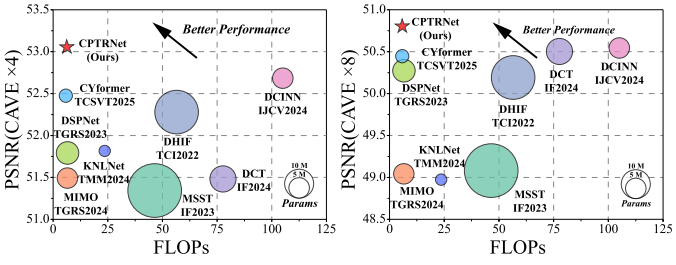


Fig. 5. Accuracy-efficiency comparison with existing methods on the CAVE dataset at the $\times 4$ and $\times 8$ scales. Models closer to the top-left corner achieve better performance. Circle size denotes the number of parameters.

first 150 epochs. The hyperparameters were set to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. The training process lasted for 600 epochs with a batch size of 16. Owing to its proven effectiveness in low-level vision tasks, we employed the ℓ_1 loss, which empirically outperformed alternative loss functions.

3) Evaluation Metrics: To evaluate fusion performance across the four simulated datasets, we used four widely adopted metrics: PSNR, SAM, ERGAS, and SSIM. We further assessed computational efficiency in terms of parameter count (Params), floating-point operations (FLOPs), GPU memory usage, and inference time. For downstream hyperspectral image classification, we reported class-specific accuracy for six land-cover categories (C1 to C6), together with overall accuracy (OA), average accuracy (AA), and the kappa coefficient.

C. Comparison with State-of-the-Art Methods

To evaluate the performance of CPTRNet, we compare it against fourteen representative baselines: three traditional model-based methods, two unsupervised deep learning-based methods, and nine supervised deep learning-based methods. The traditional model-based methods are HySure [42], GTNN [43], and GNLTN [44]. The unsupervised deep learning-based methods are FusGAT [31] and DTUML [30]. The supervised deep learning-based methods are DTLP [24], DHIF [45], MSST [46], DSPNet [47], MIMO [48], KNLNet [49], DCINN [51], DCT [50], and CYformer [52].

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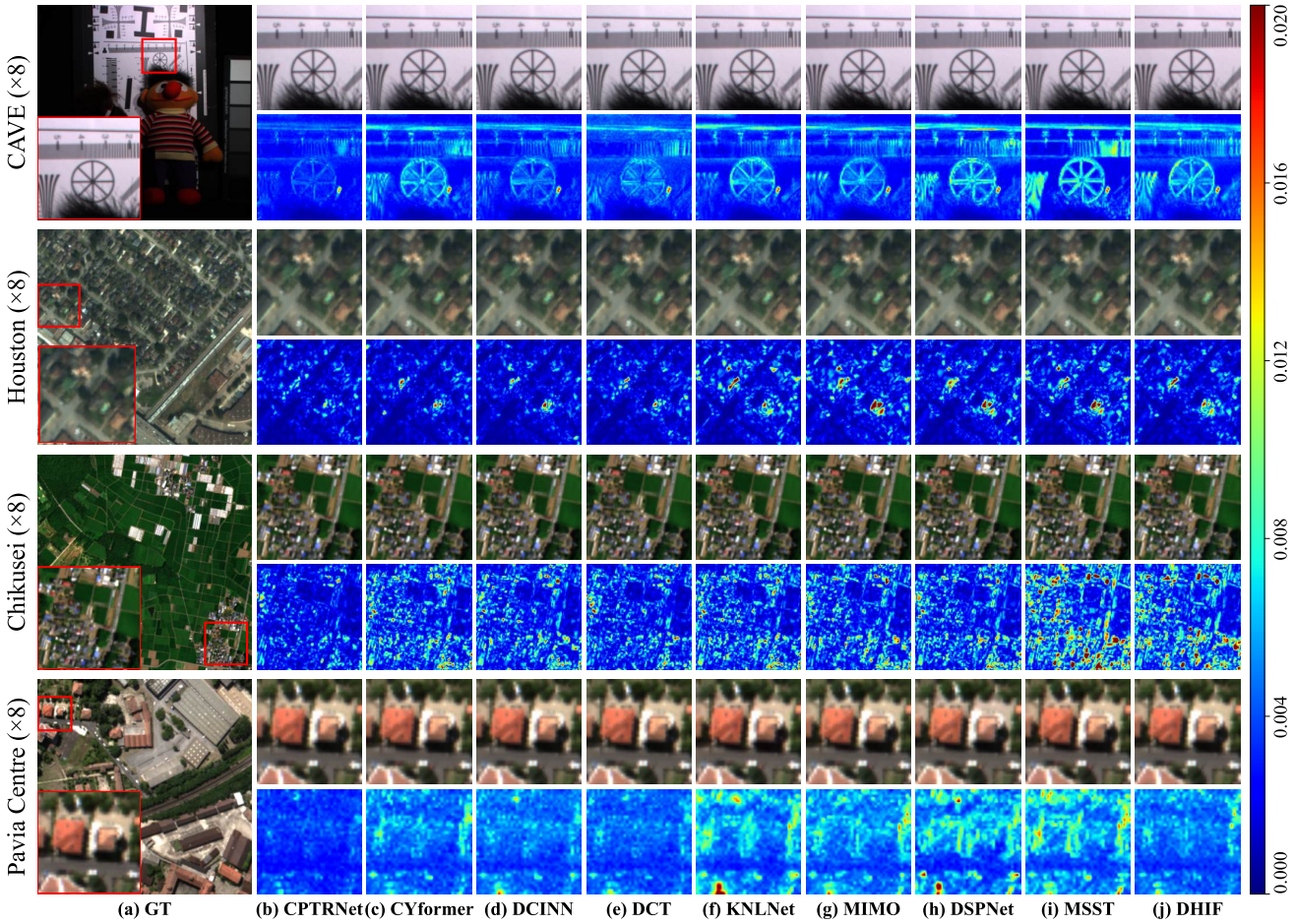


Fig. 6. Visual comparisons of representative methods on four datasets at the $\times 8$ scale. The four large images depict the pseudo-color outputs for representative samples from each dataset, with red rectangles highlighting locally enlarged regions. The odd-numbered rows show locally magnified pseudo-color visualizations produced by each method, whereas the even-numbered rows show the corresponding residual maps between the GT and the respective method outputs.

TABLE II
WALL-CLOCK INFERENCE TIME AND GPU MEMORY USAGE OF DIFFERENT METHODS ON THE CAVE ($\times 8$) DATASET. THE BEST AND SECOND-BEST VALUES ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Metric	DTLP	DHIF	MSST	DSPNet	MIMO	KNLNet	DCT	DCINN	CYformer	CPTRNet
GPU memory (MB)↓	5758	4196	1554	3286	1336	**1060**	2330	1364	2522	<u>1230</u>
Inference time (s/img)↓	2.717	1.958	1.942	2.058	3.608	**1.575**	5.450	3.658	2.508	<u>1.675</u>

1) Quantitative Comparison: Table I reports the comparison results between the proposed method and the fourteen competing methods. The results show that CPTRNet outperforms all competing methods across all metrics on the four datasets. For instance, in terms of PSNR, our method surpasses the second-best approach by 0.26, 0.45, 0.50, and 0.42 dB on the CAVE, Houston, Chikusei, and Pavia Centre datasets, respectively, under the $\times 8$ setting. Moreover, compared with the second-best method, CPTRNet reduces the SAM value by 0.084 on CAVE, by 0.084 on Houston, by 0.103 on Chikusei, and by 0.042 on Pavia Centre under the $\times 8$ setting. This clearly indicates that CPTRNet effectively preserves spectral features and significantly reduces spectral distortion. Notably, CPTRNet shows a more pronounced performance advantage on complex datasets such as Chikusei and Pavia Centre, which suggests that the combination of GKConv and CP-based redundancy reduction is particularly beneficial in these settings. In addition to its superior reconstruction performance, CPTRNet also stands out for its efficiency. The model contains only 1.43 million parameters and requires only 3.99 GFLOPs on the CAVE dataset under the $\times 8$ setting, thereby maintaining a compact model size and low computational cost. Fig. 5 further illustrates this accuracy-efficiency trade-off on the CAVE dataset at the $\times 4$ and $\times 8$ scales. CPTRNet is located in the upper-left region of both plots, indicating high reconstruction accuracy with low computational cost. The hardware benchmarks in Table II corroborate the practical efficiency of CPTRNet. The results show that CPTRNet achieves the second-best performance in both GPU memory consumption and wall-clock inference time. While KNLNet is slightly faster due to its emphasis on lightweight execution, CPTRNet maintains a better balance between computational

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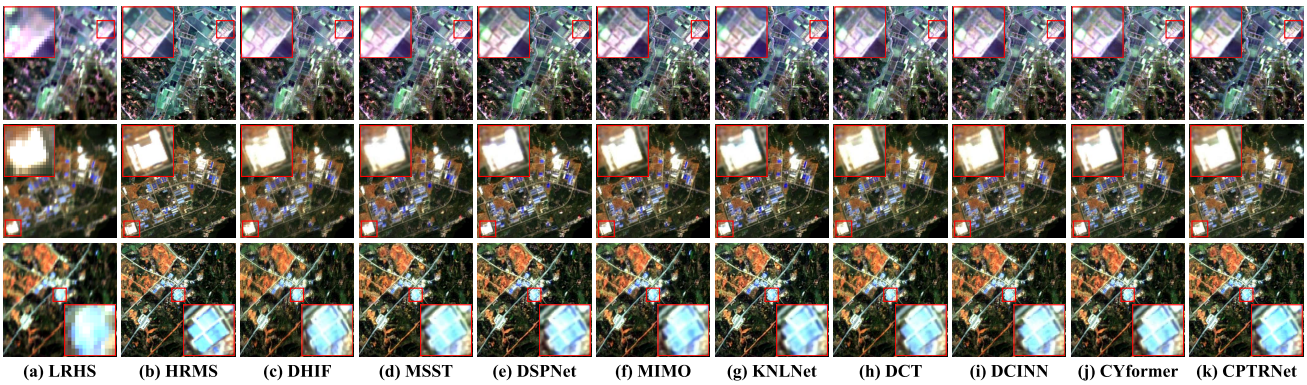


Fig. 7. Visual comparisons on the real-world WHU-MHF dataset. Three representative scenes are presented, including a rural field scene, an industrial-area scene, and a village scene. The first two columns show the original LRHS (30-m GSD, 148 bands) and HRMS (10-m GSD, 8 bands) observations, respectively, while the remaining columns show the pseudo-color fusion results produced by different methods. The red boxes indicate locally enlarged regions for detailed comparison.

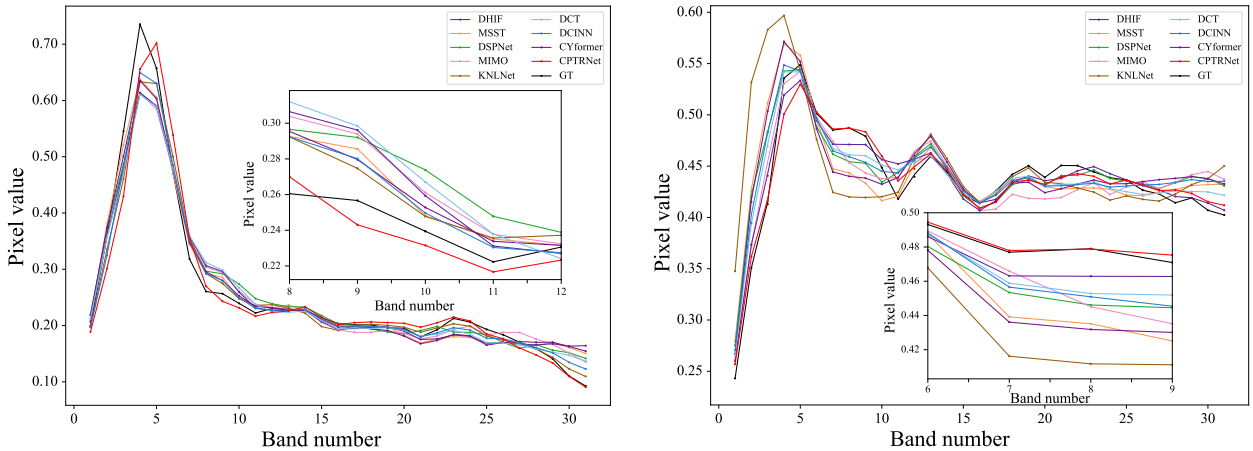


Fig. 8. Comparison of the spectral consistency of the proposed method with that of state-of-the-art methods. (a) Spectral vectors of *stuffed toys* at pixel location (97, 156); (b) Spectral vectors of *thread spools* at pixel location (131, 207).

efficiency and reconstruction fidelity. These hardware-level results confirm the practical viability of our framework for real-world remote sensing applications.

2) Qualitative Comparison: Fig. 6 presents representative results on the CAVE, Houston, Chikusei, and Pavia Centre datasets. Compared with competing methods, CPTRNet demonstrates clear visual advantages, particularly in preserving fine texture details while introducing fewer artifacts. In the magnified regions, the residual maps further reveal smaller reconstruction errors for CPTRNet, indicating superior reconstruction quality. On the CAVE dataset, the proposed method effectively exploits both spatial and spectral information to reduce blur artifacts and preserve crisp contours. On the Houston dataset, the proposed method reconstructs complex urban patterns with clearer building outlines and road boundaries, while better preserving structural details in densely packed residential areas. On the Chikusei dataset, CPTRNet accurately captures subtle spatial structures and better preserves the boundaries of houses and fields, thereby alleviating blurring and over-smoothing. In the urban Pavia Centre dataset, CPTRNet reconstructs architectural features such as rooftops, roadways, and courtyards with high fidelity, while suppressing common artifacts and producing clearer building edges and roof boundaries than competing methods.

To further evaluate the generalization capability of the proposed method in real-world remote sensing scenarios, Fig. 7 presents visual comparisons of three representative scenes from the WHU-MHF dataset. In the first scene, CPTRNet preserves rural field boundaries and road structures more effectively while reducing the blurring effects observed in the competing methods. In the second scene, the proposed method reconstructs buildings with greater clarity, resulting in sharper contours and more distinct structural outlines. In the third scene, CPTRNet more faithfully recovers roof edges and geometric patterns in the enlarged region, yielding sharper textures and fewer visual distortions. Overall, CPTRNet produces fused images with richer spatial details and clearer structural patterns than the competing methods.

In addition, Fig. 8 presents a comparison of spectral vectors for two representative samples from the CAVE dataset, with selected band ranges enlarged for closer inspection. A closer examination reveals that the spectral vectors produced

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TABLE III
QUANTITATIVE COMPARISON OF DOWNSTREAM CLASSIFICATION RESULTS OBTAINED FROM THE RECONSTRUCTED OUTPUTS OF SELECTED REPRESENTATIVE METHODS. THE BEST AND SECOND-BEST VALUES OF THE OVERALL METRICS OA, AA, AND KAPPA ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Method	C1 (%)↑	C2 (%)↑	C3 (%)↑	C4 (%)↑	C5 (%)↑	C6 (%)↑	OA (%)↑	AA (%)↑	Kappa (%)↑
DTLP	97.64	99.38	98.77	57.83	89.74	93.30	82.76	89.44	78.77
DHIF	97.66	99.38	99.08	53.63	89.12	92.93	81.22	88.63	76.98
MSST	97.49	99.43	98.77	56.28	90.52	93.40	82.28	89.31	78.22
DSPNet	97.59	99.34	98.00	58.16	89.27	93.01	82.74	89.23	78.73
KNLNet	97.66	99.34	98.77	58.80	89.82	93.45	83.13	89.64	79.20
DCT	97.57	99.34	98.77	59.62	87.86	93.43	83.25	89.43	79.32
DCINN	97.47	99.29	98.61	58.24	90.92	93.22	82.92	89.62	78.94
CYformer	97.42	99.34	98.77	58.95	90.60	93.61	83.22	89.78	79.31
CPTRNet	97.42	99.03	97.53	65.30	89.74	93.93	**85.33**	**90.49**	**81.74**

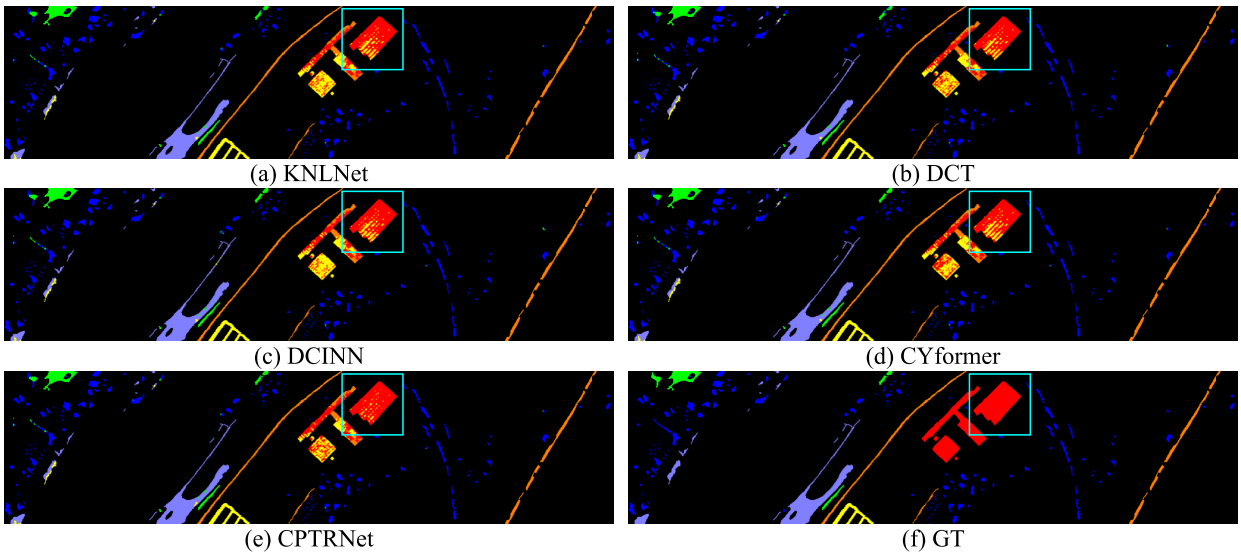


Fig. 9. Visual classification maps for the downstream classification task generated from the reconstructed results on the Pavia Centre dataset. Blue rectangles highlight local regions with noticeable variations to facilitate clearer comparison.

by CPTRNet closely align with the GT, demonstrating the model's ability to faithfully reconstruct the original spectral characteristics of the images and accurately preserve fine details and subtle variations in the spectral domain. Overall, the quantitative and qualitative results on the four simulated datasets, together with the visual results on the real-world WHU-MHF dataset, confirm the effectiveness of the proposed method for HSI-MSI fusion and demonstrate its clear performance advantages over existing techniques.

D. Impact on Downstream Classification

An important application of HSI-MSI fusion is to facilitate downstream analytical tasks, such as hyperspectral image classification. To evaluate the spatial-spectral fidelity from this perspective, we conducted a comparative classification study by applying a support vector machine (SVM) [58] to the reconstructed outputs of selected representative methods on the Pavia Centre dataset. To ensure a fair comparison, the same SVM training and testing samples were used for the reconstructed outputs of selected representative methods. As summarized in Table III, the proposed CPTRNet achieves the best OA, AA, and kappa coefficient among the compared methods. These improvements are visually corroborated in Fig. 9, where blue rectangles highlight specific local regions with noticeable variations. Consistent with the quantitative findings, the classification maps generated from the CPTRNet reconstructions exhibit higher spatial consistency and fewer misclassified pixels.

E. Ablation and Analysis

1) Low-Rank Analysis of CAVE HSIs: To examine the intrinsic spectral redundancy of the CAVE dataset, we computed the cumulative singular-value energy and inter-band correlation statistics using all original HSIs in the dataset. As shown in Fig. 10, a small number of dominant components capture most of the spectral variance, while strong correlations are observed among spectral bands. These results provide empirical support for the low-rank characteristics commonly observed in HSIs.

2) Ablation of Crucial Components: To assess the contribution of each module to the overall model performance, we conducted ablation experiments on different variants of

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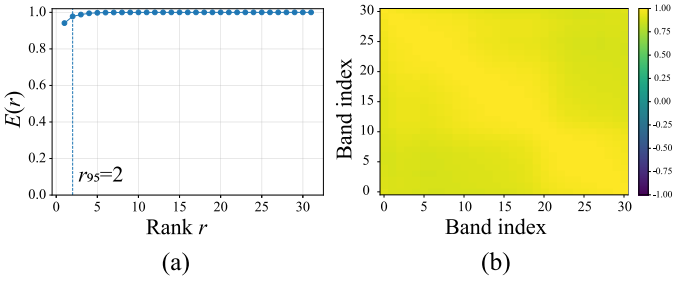


Fig. 10. Spectral statistics averaged over all images in the CAVE dataset. (a) Cumulative energy $E(r)$ as a function of truncation rank r. The marker r_{95} indicates the smallest rank at which $E(r) \geq 0.95$. The rapid increase and early saturation of the curve suggest that only a small number of components account for the majority of spectral variance, reflecting the low-rank nature of HSIs. (b) Inter-band correlation heatmap computed across all pixels. The dominance of near-unit correlations reveals strong spectral redundancy and collinearity, consistent with the low-rank structure of hyperspectral data.

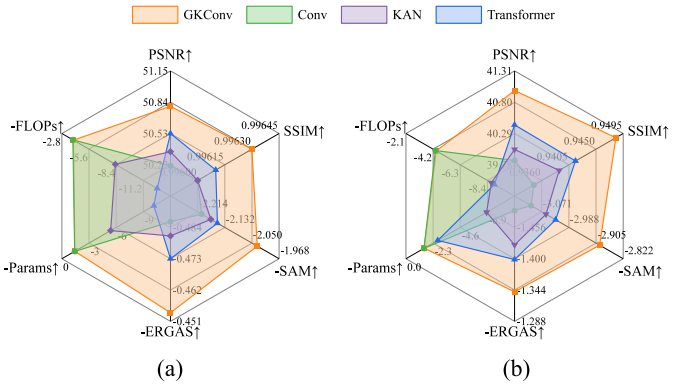


Fig. 11. Ablation study of GKConv on (a) CAVE and (b) Chikusei at the $\times 8$ scale. Metrics for which lower values indicate better performance were negated so that larger radial values indicate better performance on all axes.

TABLE IV
ABLATION STUDY OF KEY MODEL COMPONENTS ON THE CAVE DATASET AT THE $\times 8$ SCALE. THE BEST AND SECOND-BEST VALUES ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Variant	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params↓	FLOPs↓
w/o GKConv & KVG	49.96	0.9958	2.259	0.507	**1.30**	**3.92**
w/o KVG	50.54	0.9962	2.128	0.474	1.31	3.98
w/o GKConv	50.21	0.9960	2.202	0.486	1.41	3.93
w/o CP	50.42	0.9961	2.195	0.476	1.43	3.99
CPTRNet	**50.80**	**0.9963**	**2.034**	**0.454**	1.43	3.99

TABLE V
ABLATION STUDY OF DIFFERENT BASIS FUNCTIONS ON THE CAVE DATASET AT THE $\times 8$ SCALE. THE BEST AND SECOND-BEST VALUES ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Basis Function	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params↓	FLOPs↓
B-spline	50.08	0.9959	2.185	0.494	6.72	8.25
RBF	50.26	0.9960	2.167	0.484	3.35	6.75
ReLU	50.13	0.9959	2.173	0.493	2.24	5.74
Wavelet	50.52	0.9961	2.135	0.471	1.75	12.64
Gram polynomial	**50.80**	**0.9963**	**2.034**	**0.454**	**1.43**	**3.99**

TABLE VI
ABLATION STUDY OF THE CP RANK HYPERPARAMETER ON THE CAVE DATASET AT THE $\times 8$ SCALE. THE BEST AND SECOND-BEST VALUES ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Setting	Value	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Params↓	FLOPs↓
CP rank r	1	49.43	0.9952	2.443	0.587	**0.83**	**1.86**
	2	50.22	0.9959	2.383	0.494	1.13	2.92
	3	**50.80**	**0.9963**	**2.034**	**0.454**	1.43	3.99
	4	50.74	0.9962	2.083	0.467	1.72	5.05
	5	50.72	0.9961	2.088	0.475	2.02	6.12

CPTRNet. The experimental results on the CAVE dataset are presented in Table IV. For the ablation variants, GKConv, KVG, and the CP outer-product operation were replaced with a standard 3×3 convolution, global average pooling, and additive fusion, respectively. The results indicate that the introduction of the GKConv and KVG modules significantly improves performance, confirming their effectiveness.

3) Ablation of GKConv: To validate the effectiveness of GKConv in RTG, we conducted separate experiments in which GKConv was replaced by the original KAN, a traditional convolution, or a Transformer-based module. As shown in Fig. 11, traditional convolutions incur lower computational overhead but fail to maintain competitive reconstruction performance. In contrast, the original KAN and Transformer-based modules provide more powerful feature representations at the expense of higher computational costs. GKConv achieves a favorable balance between efficiency and accuracy, delivering robust reconstruction quality at relatively low computational overhead.

4) Ablation of Basis Functions: To evaluate the performance of different basis functions in KAN convolution, we conducted an ablation experiment on the CAVE dataset, as summarized in Table V. Although radial basis functions (RBFs) and ReLU basis functions reduce computational overhead compared with B-spline basis functions because of their simpler operations, they do not yield significant performance gains. Wavelet basis functions achieve strong performance because of their precise image representation capability, but at a substantially higher computational cost. In contrast, Gram polynomials provide strong approximation capability for high-dimensional functions while maintaining low computational cost, owing to their orthogonality and numerical stability.

5) Hyperparameter Settings: To further investigate the impact of the CP rank r on model performance, we evaluated the model with $r \in \{1, 2, 3, 4, 5\}$. The results on the CAVE dataset are shown in Table VI, demonstrating that the model achieves the best performance among the tested settings when $r = 3$, which is therefore adopted as the default setting.

V. CONCLUSION

In this paper, we propose CPTRNet, a framework for HSI-MSI fusion that integrates low-rank tensor priors into a fully differentiable architecture. Through the RTG module, CPTRNet employs GKConv to extract expressive nonlinear features from multimodal fused representations. KVG then aggregates these features into three mode-wise factor vectors, whose outer product forms a rank-1 tensor. By progressively generating and

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aggregating multiple rank-1 components, CPTRNet achieves structured low-rank tensor reconstruction in an end-to-end, iteration-free manner. Experiments on five datasets demonstrate that our approach outperforms existing methods while requiring fewer parameters and lower computational cost.

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